

Demonstration of CoilCAM: an Action-Oriented Toolpath Programming System for Clay 3D Printing

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Figure 1: Clay 3D printed tea scene. A) Schematics of the toolpath operations required to create the teapot with CoilCAM. We use a Boolean union operator to combine the body and the spout and two Boolean difference operators to produce generate the handle. B) Tea set in function with 3D printed plant pot demonstrating CoilCAM's ability to create surface textures.

ABSTRACT

Clay 3D printing provides the benefits of digital fabrication automation and reconfigurability through a method that evokes manual clay coiling. Existing design technologies for clay 3D printing reflect the general 3D printing workflow in which solid forms are designed in CAD and then converted to a toolpath. In contrast, in hand-coiling, form is determined by the *actions* taken by the artist's hands through space in response to the material. We theorized that an *action-oriented* approach for clay 3D printing could allow creators to design digital fabrication toolpaths that reflect clay material properties. We present CoilCAM, a domain-specific CAM programming system that supports the integrated generation of parametric forms and surface textures through mathematically defined toolpath operations.

CCS CONCEPTS

• **Human-centered computing** → *Interactive systems and tools*.

KEYWORDS

Digital Fabrication; Computer-Aided Machining; Clay 3D Printing; Artist Residency

ACM Reference Format:

Samuelle Bourgault, Devon Frost, Pilar Wiley, Avi Farber, and Jennifer Jacobs. 2023. Demonstration of CoilCAM: an Action-Oriented Toolpath Programming System for Clay 3D Printing. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3544549.3583921>

1 INTRODUCTION

The 4,000-year-old practice of clay coiling, wherein craftspeople build vessels layer by layer with hand-rolled coils [4], exemplifies many key qualities of manual craft. Hand-coiling involves a non-hylomorphic workflow [8] where form is the result of a dialogue between the craftspeople's hands and the material. In this process, design and fabrication tasks happen simultaneously. Each manual operation impacts the final outcome and the artifact qualities are in flux and at risk throughout production [12]. The integration of

design and fabrication in coiling is powerful because this process allows the craftsperson to experience material forces and modify their design in response.

Despite its ancient origins, hand-coiling is also aligned with emerging digital methods of clay fabrication. Similar to coiling, clay 3D printers fabricate 3D structures by extruding coils of clay, which are vertically layered and compacted in succession [17]. Clay 3D printing can extend clay production through the precision, repeatability, and programmable nature of computer-numerical-control (CNC) fabrication. Using clay 3D printers, craftspeople can manufacture physical forms that are, in part, defined by a digital model [9]. While clay 3D printers use similar materials and concepts as hand-coiling, they require fundamentally different tools. Clay 3D printing often involves a workflow that is similar to other forms of digital fabrication wherein craftspeople develop a geometry in computer-aided design (CAD) software and convert the geometry into machine instructions using computer-aided manufacturing (CAM) software [16].

Circumventing CAD and designing exclusively in CAM offers unique opportunities for clay 3D printing. In a CAM-based design workflow, geometries are defined by specifying toolpath operations to be executed by the CNC machine rather than by automatically slicing solid digital models. CAM-based design requires extensive knowledge of the relationship between the machine and the material but affords a greater range of material expressiveness [10] by allowing the craftsperson to specify fabrication behaviors that take advantage of material properties [11]. These opportunities are particularly salient for the intersection of manual clay craft and digital fabrication. Ceramic craftspeople have extensive material knowledge and clay supports many different surface qualities and forms depending on how it is worked. To take advantage of the versatility of clay, clay 3D printing practitioners often use general-purpose programming tools to create custom CAM-based design workflows that produce unique and material-specific outcomes [3, 6, 15].

We observed parallels between CAM-based design and hand-coiling techniques. In both cases, form emerges from the interaction between each action enacted by the craftsperson or machine and the material response. We introduce the term **action-oriented** to characterize manual and machine fabrication workflows in which form is conceived of through iterative manual or machine actions. We contrast action-oriented workflows with **artifact-oriented** workflows, in which generalizable fabrication operations are applied to a material-agnostic design – for example using a general-purpose slicing algorithm to create toolpaths from a solid CAD model. Our definition of action-oriented workflows builds on existing models of human-machine collaboration—namely Devendorf and Rosner’s model of “coproductions” [5] and Andersen *et al.*’s concept of “digital craft-machine-ship” [1]. However, our focus is to categorize incremental actions that occur between material, manual or machine agents in the fabrication process, which, in turn, can directly inform new fabrication system abstractions.

We theorized that an action-oriented clay 3D printing system could enable craftspeople to design and fabricate both forms and surface textures while harnessing material-specific. To explore this theory, we developed **CoilCAM**, a domain-specific visual programming system that supports CAM-based design for clay 3D printing. CoilCAM includes a set of CAM abstractions—toolpath generators,

mathematical shaping functions, and Boolean operators—that support the design of 3D-printed ceramics by authoring 3D printing machine toolpaths. We developed CoilCAM as a collaboration between two HCI researchers and two professional ceramic craftspeople. We structured our collaboration within a 10-week residency that followed a Research through Design (RtD) approach. We evaluated CoilCAM by reinterpreting three pieces originally created by hand by our craft collaborators. Our research builds on a growing body of work exploring the benefits of CAM-based design for additive thermoplastic fabrication [7, 11, 13, 14]. Unlike prior work, we contribute a *clay-specific* CAM-based design system and evaluate our approach with respect to techniques and outcomes in manual ceramics craft. This work describes the interactive demonstration of the CoilCAM system, presented in the full paper by Bourgault *et al.* [2].

2 SYSTEM DESCRIPTION

CoilCAM is a visual programming system that enables the creation of mathematically defined toolpaths to create 3D-printed ceramics. In this section, we describe the CoilCAM system primitives.

2.1 System Description

We developed CoilCAM as a visual programming system using Python Script components in the visual programming language Grasshopper.¹ Figure 2 shows an implemented CoilCAM program to make the vessel depicted in Figure 5B. Programming in CoilCAM is centered around the use of toolpath primitives that can be modified with a collection of mathematical operators. By combining operators in different ways, the craftsperson can generate a wide variety of complex toolpaths. These toolpaths are then sent to a clay 3D printer to be executed. In this extended abstract, we represent CoilCAM programs as a series of diagrams with color-coded nodes that correspond to our custom Grasshopper components. Some of the ceramic pieces shown in the following figures have been decorated with glazes, underglazes, wood-firing, and pit-firing. These decorations are not a feature of CoilCAM but show how artwork made with our system can be decorated using similar techniques to hand-made artwork. In the remainder of this section, we describe the different components of the system.

2.2 Toolpath Unit Generator and Function Operators

Programming in CoilCAM is structured around procedural transformations of a cylindrical toolpath represented by the *toolpath unit generator* component. The default toolpath unit generator, represented as a green module in Figure 3, generates a 2D circular toolpath on the xy-plane comprised of a series of points along a circle. This circular toolpath is repeated to produce a 3D toolpath as a series of layers with increasing height along the z-axis.

The toolpath unit generator contains **five** initialization parameters—position, initial radius, layer height, number of two-dimensional layers, and number of points along the layer (not represented on the toolpath unit generator module but listed in Figure 3) and **four** shaping parameters—radius, scale, translate and rotate (represented

¹To download a functioning version of CoilCAM, visit <http://coilcam.com>

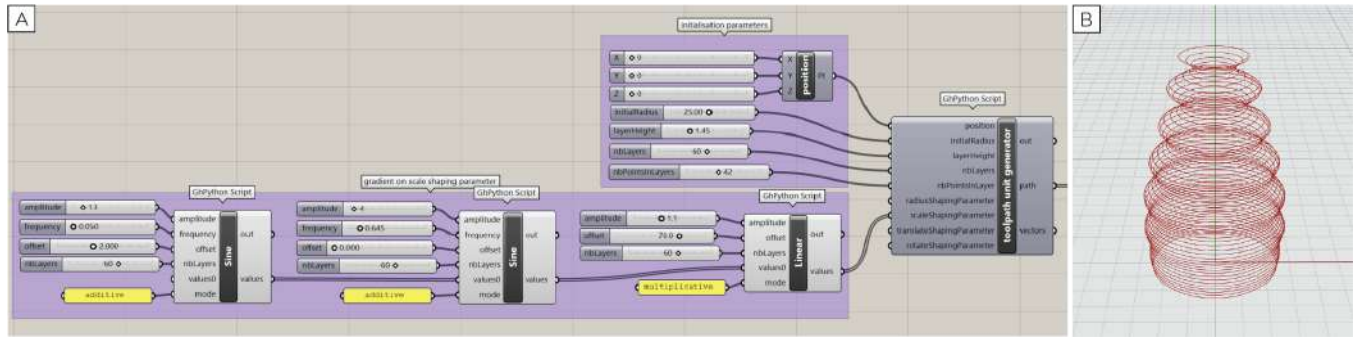


Figure 2: CoilCAM workflow in Grasshopper to make Figure 5B. A) The toolpath unit generator with a linked set of function operators acting on the scaling shaping parameter and creating a gradient on the sinusoidal amplitude. B) The toolpath representation of that workflow.

as input on the toolpath unit generator on Figure 3). The radius shaping parameter controls the shape of each layer by specifying the polar coordinate of each point along the toolpath at that layer. The scale, translate and rotate shaping parameters shape the toolpath by respectively scaling, translating, or rotating individual layers in succession.

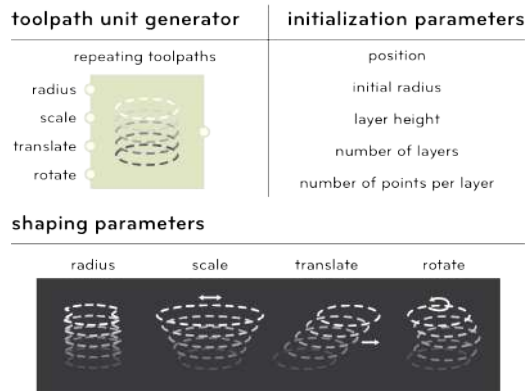
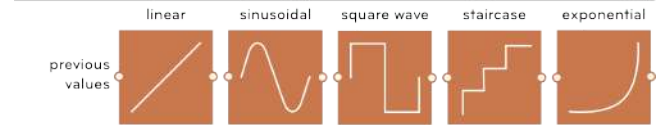


Figure 3: The toolpath unit generator and its initialization and shaping parameters. The default generator consists of a series of two-dimensional layers comprised of a set of points in a circular path. The generator is initialized with a position, a radius, a layer height, a number of layers, and a number of points in a layer. The generator contains four shaping parameters: radius, scale, translate and rotate. By connecting functions into the shaping parameters, the craftsperperson can respectively create non-circular layers, scale up layers to create flared openings, translate layers to generate shifted forms, and rotate layers to make twisting forms.

The craftsperperson can modify the initialization parameters' values with a slider to set a different constant. The shaping parameters can be modified in an iterative fashion by setting their input to a *function operator*: a component that returns a series of values corresponding to a predefined mathematical function. As shown in Figure 4, the system contains five function operators: linear, sinusoidal, square wave, staircase, and exponential. Function operators

can be mapped as input to any toolpath unit generator shaping parameter. Alternatively, they can be chained in sequence to modify the output of a prior function operator. Each function operator can be used in two different modes that determine its behavior when connected to another component: an additive mode which increments the previous input values and a multiplicative mode which multiplies the previous input values. We represent schematically the operation mode using + and × over the connections between function operators.

function operators



Boolean operators

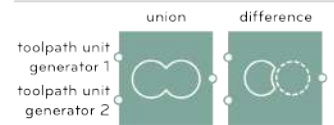


Figure 4: The system has two types of operators: function and Boolean operators. Function operators are used to define the toolpath unit generator's shaping parameters. Boolean operators are used to merge or subtract toolpath unit generators.

The modularity of these function operators enables the craftsperperson to develop a variety of forms and textures by chaining function operators together in different sequences. For instance, Figure 5A shows how we can create a sinusoidal gradient on the radius shaping parameter by linking a sinusoidal function operator with a linear function operator in multiplicative mode. Figure 5B illustrates how the same function operators can yield widely different results when applied to the scale shaping parameter instead of the radius shaping parameter represented in Figure 5A.

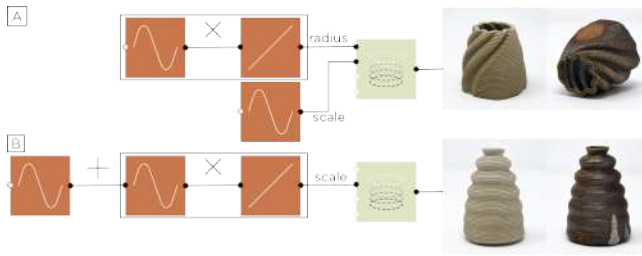


Figure 5: Gradient effect applied by chaining function operators. A first sinusoidal function operator defines the inward shape on the scale shaping parameter of both pieces. A second sinusoidal function operator is multiplied with a linear function operator to generate a linear gradient over the sinusoidal amplitude, which modulates A) the radius and B) the scale of the object.

2.3 Boolean Operators

Existing CAD tools for clay 3D printing are largely limited to the design and fabrication of cylindrical geometries. To support non-cylindrical forms, CoilCAM includes *union and difference Boolean operators*. As described in Figure 4, Boolean operators accept two toolpath unit generators as input and perform either a union or difference operation between each toolpath of the first input and each corresponding toolpath of the second input. Each Boolean operator outputs a toolpath unit generator with values that reflect the Boolean operation. Figure 6A and B show the process of using Boolean union and Boolean difference operators in CoilCAM and the resulting artifacts.

3 DEMONSTRATION STRUCTURE

For our interactivity session, we present a demonstration of the CoilCAM system through an interactive demonstration of the CoilCAM software and a corresponding live printing of clay vessels on a small PotterBot 3D printer. The live clay 3D printing demonstrates the effectiveness of CoilCAM as a design system and introduces a wider audience to the process and challenges involved in clay 3D printing. Our demonstration also includes a series of finished and fired artifacts created with CoilCAM. We display completed glazed works produced with CoilCAM to demonstrate the wide variety of forms and textures enabled by our system. We also display an iterative sequence of seven vessels with corresponding block diagrams showing the steps of chaining functions to create a final product—a pitcher. Figure 7 shows the sequence of vessels, which helps illustrate the iterative design process enabled by our system. Visitors are able to handle the vessels to assess the structural and aesthetic qualities of designs made entirely with CoilCAM.

ACKNOWLEDGMENTS

We are grateful for the help of our fellow Expressive Computation Lab members at UCSB, especially Ph.D. Candidate Mert Toka; and the Clay Studio in Santa Barbara. We also appreciate the support of our collaborators in the Hand and Machine Lab, Tim Castillo, Manuel Montoya, and Ted Jojola at UNM. Finally, we wish to thank

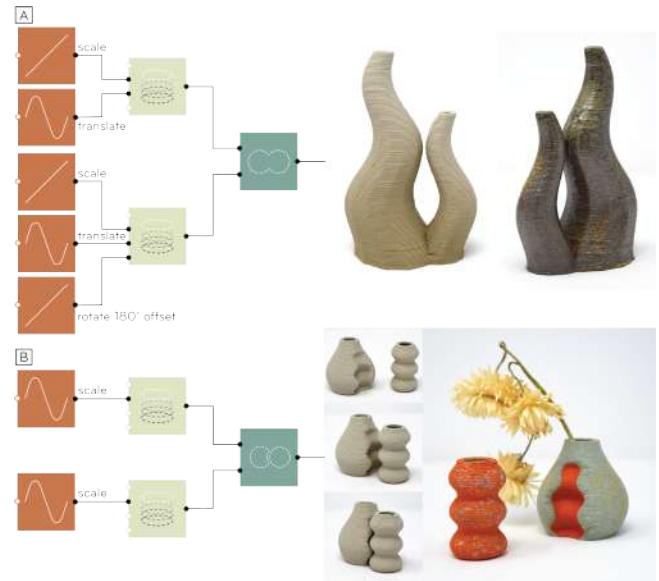


Figure 6: Using the Boolean operators in CoilCAM. A) Two toolpath unit generators are created by applying a linear function operator to their radius shaping parameter and a sinusoidal function operator to their translate shaping parameter. A linear function operator of 0 amplitude and 180° offset is applied to the rotation of the second generator to align the seam inward of the vessel. The two toolpath unit generators are then combined using a Boolean union operator and result in a two-headed wavy structure. B) Two toolpath unit generators are shaped using a sinusoidal operator on the scale shaping parameter and combined using a Boolean difference operator. This approach creates complementary vessels that can fit into one another.

Laura Devendorf, Shelby Doyle, and Sarah Rosalena Brady for guidance in our residency structure. This research was funded in part by the NSF IIS Future of Work at the Human-Technology Frontier Program (Award: 2026286) and the NSF IIS Human-Centered Computing Program (Award: 2007094).

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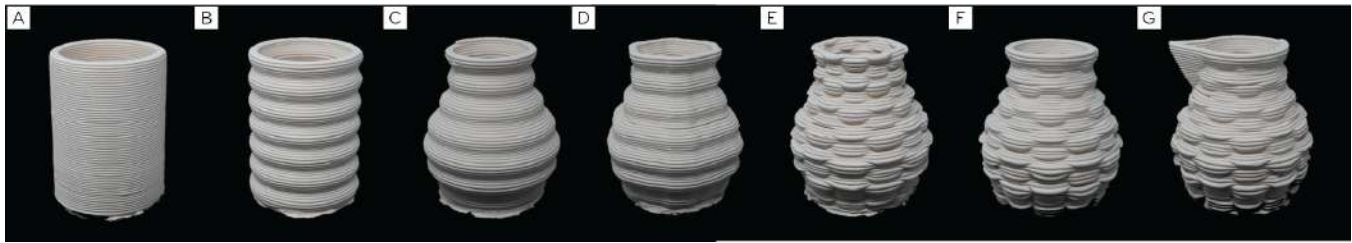


Figure 7: Step-by-step process of the design of a textured pitcher. Each intermediary artifact involves one additional operator from the precedent phase. A) Basic unit toolpath generator. B) Sinusoidal function operator on scale shaping parameter. C) Additional sinusoidal function operator on scale. D) Square wave function operator on radius to create bumps. E) Staircase function operator on rotation to offset bumps. F) Linear function operator on scale to create a gradient. G) Boolean union operator to add a beak to the pitcher.

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